

10 DATA ENGINEERING CONCEPTS EVERY DATA ENGINEER MUST KNOW



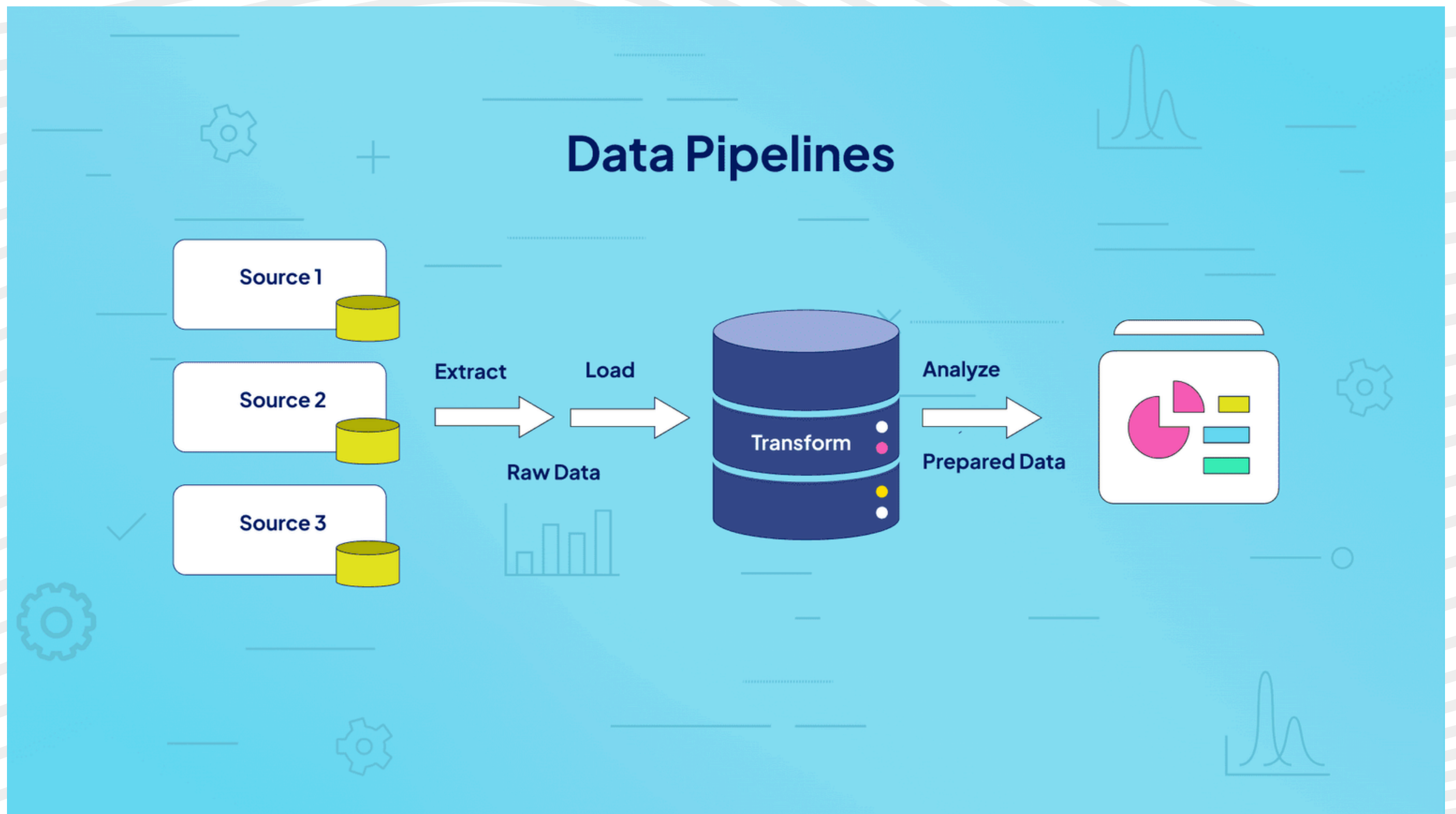
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DATA PIPELINES

A series of processes that move data from source systems to storage and analytics platforms.

Why it matters

- Automates data movement
- Ensures reliable data delivery
- Powers analytics and machine learning



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ETL(EXTRACT-TRANSFORM-LOAD)

Data is extracted from source systems, transformed in a processing engine, and then loaded into the data warehouse.

Step 1: Extract

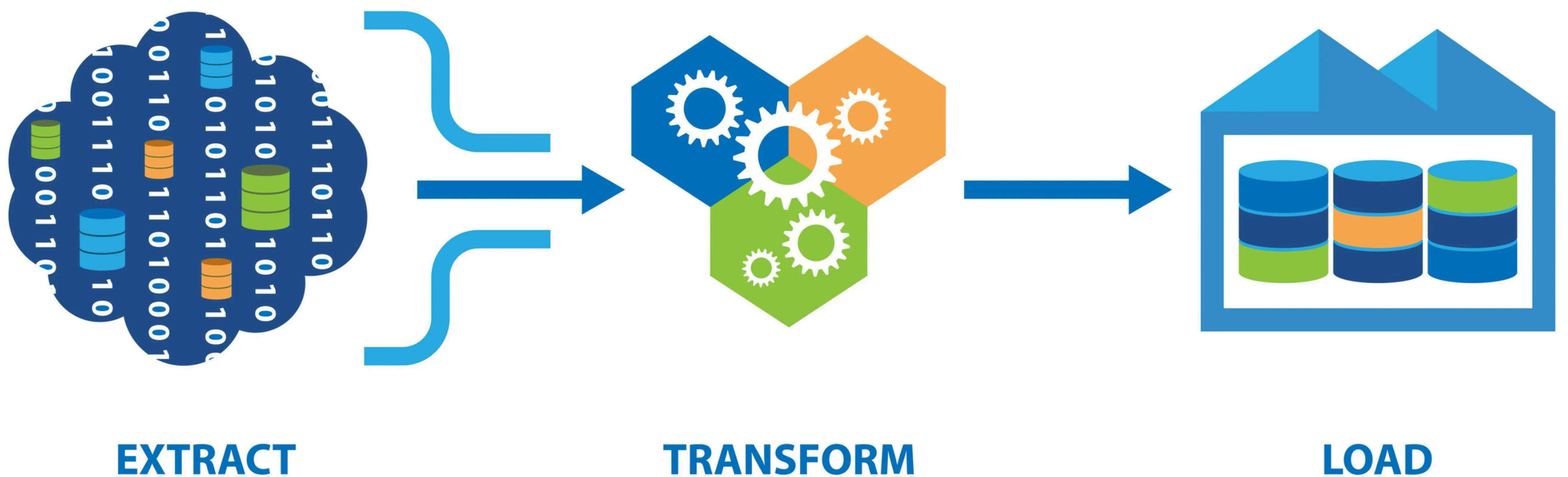
Collect data from databases, APIs, files, or applications.

Step 2: Transform

Clean, validate, enrich, and aggregate data.

Step 3: Load

Store processed data in the warehouse.



ELT- EXTRACT-LOAD-TRANSFORM

Raw data is first loaded into the data warehouse or lakehouse, then transformed using the platform's compute power.

Step 1: Extract

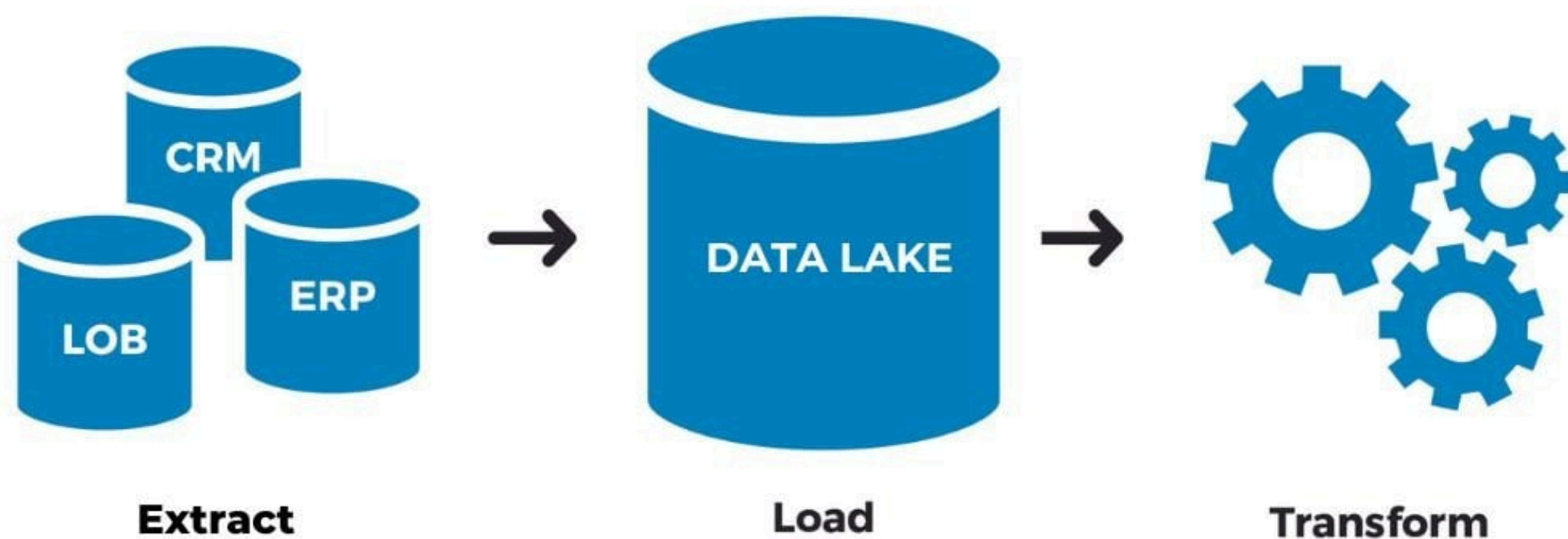
Pull data from source systems.

Step 2: Load

Store raw data directly in the warehouse.

Step 3: Transform

Use SQL and warehouse compute resources to transform data.



BATCH VS REAL-TIME PROCESSING

Real-Time Processing
Processes events
immediately as they
occur.

Examples

- Fraud detection
- Live dashboards
- Ride-sharing apps
- Recommendation engines

Batch Processing
Processes large volumes
of data at scheduled
intervals.

Examples

- Daily sales reports
- Monthly financial reports
- Data warehouse refreshes

Real-Time Operating Systems

Reacts to data in
seconds or
milliseconds

Difficult to
implement

Batch Processing

Processes large
volume of data all
at once

May require
dedicated staff to
handle issues

DATA WAREHOUSES

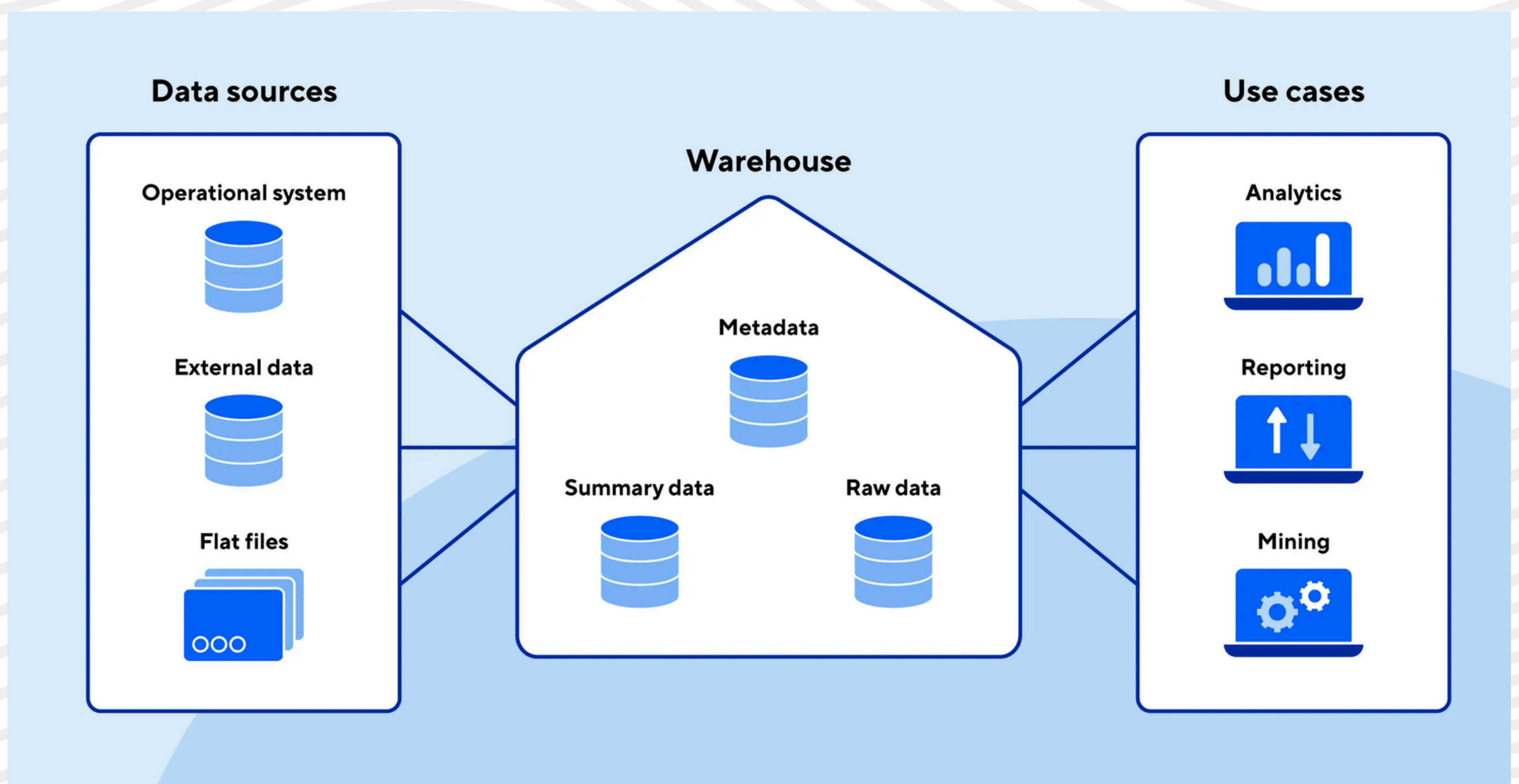
A centralized analytical database designed for reporting, business intelligence, and decision-making

Characteristics

- Structured data
- Fast SQL performance
- Historical data storage
- Optimized analytics

Popular Platforms

- Snowflake
- Amazon Redshift
- Google BigQuery
- Azure Synapse



DATA LAKES

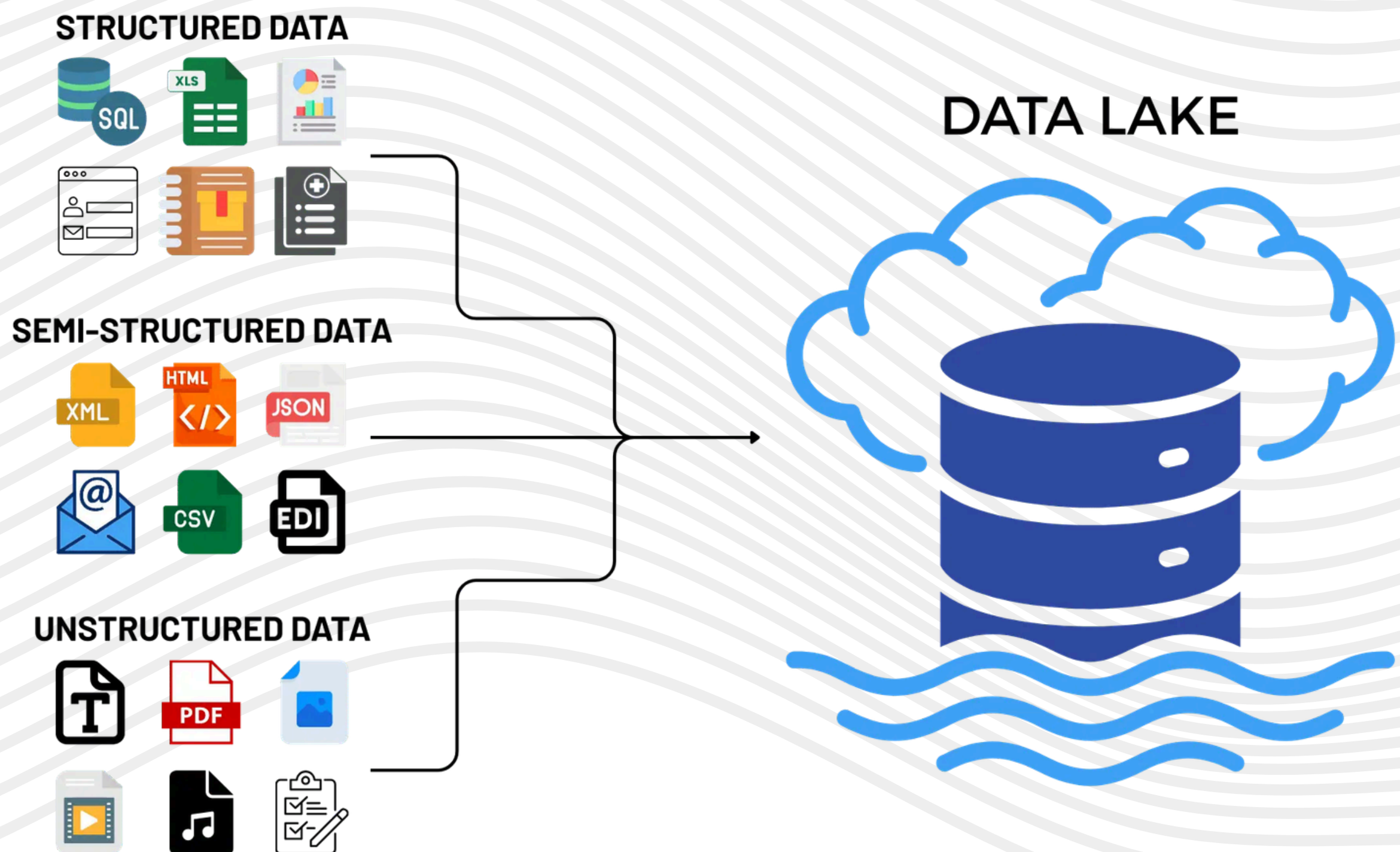
A storage repository that stores data in its original format without requiring predefined schemas.

It Provides

- Low-cost storage
- Massive scalability
- Supports AI and ML
- Flexible schema design

Common Technologies

- Amazon S3
- Azure Data Lake Storage
- Google Cloud Storage



DATA LAKEHOUSE

A modern architecture that combines the scalability of data lakes with the reliability of data warehouses.

Solves Problems Of
Data Lakes:

- Poor governance
- Slow analytics

Data Warehouses:

- High storage costs
- Limited flexibility
-

Popular Technologies

- Apache Iceberg
- Delta Lake
- Apache Hudi



**Data
Warehouse**



**Data
Lake**



**Data
Lakehouse**

DATA MODELING

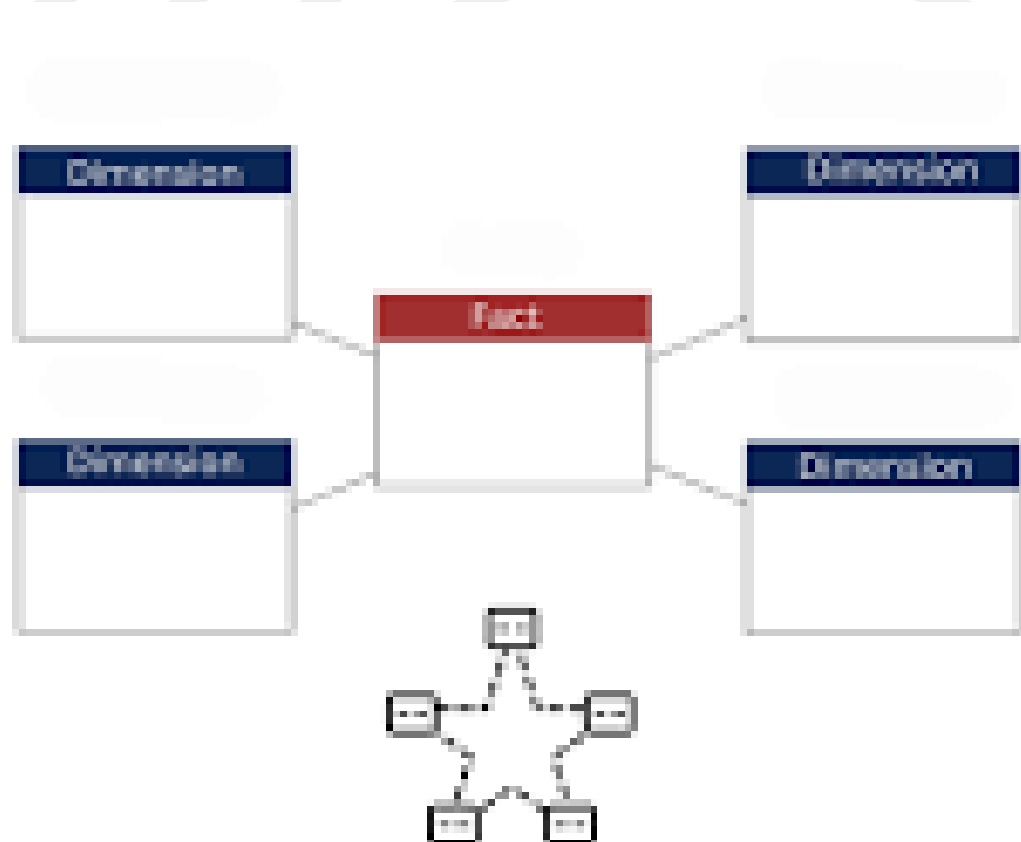
The process of organizing data into structures optimized for analytics and reporting.

Objectives

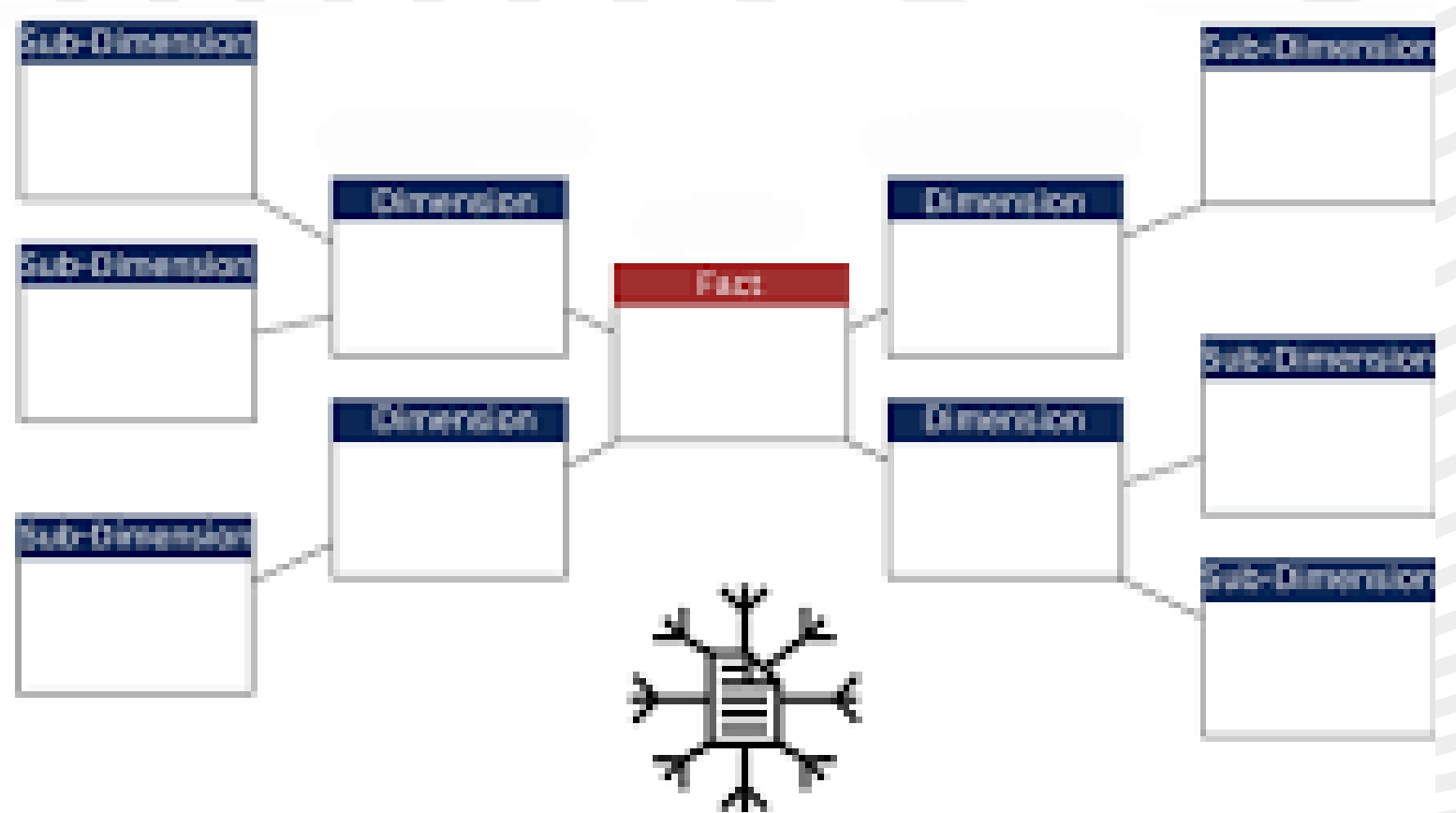
- Improve query performance
- Simplify reporting
- Ensure consistency
- Reduce complexity

It Provides better performance, easier maintenance, and more reliable analytics.

STAR SCHEMA



SNOWFLAKE SCHEMA



DATA QUALITY

Data Quality refers to the reliability, accuracy, completeness, and consistency of data used for analytics, reporting, and decision-making.

Simply put:

High-Quality Data = Trusted Insights

Poor-Quality Data = Poor Business Decisions

Example

If customer records contain duplicate entries, sales and marketing reports may overestimate customer growth.



DATA GOVERNANCE & LINEAGE

Data Governance

The framework that ensures data is secure, compliant, accurate, and properly managed.

Governance Covers

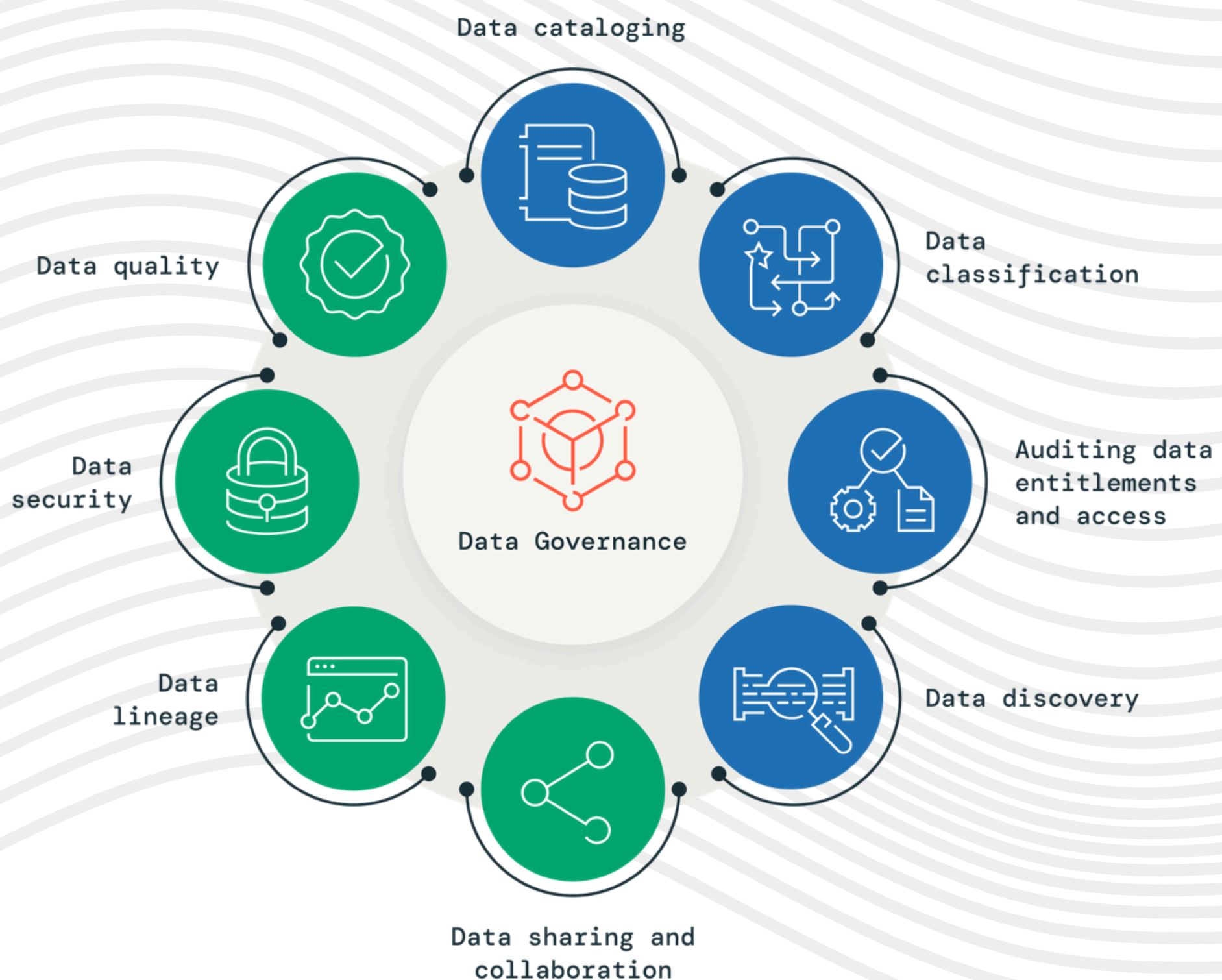
- Access Control
- Security
- Compliance
- Data Ownership
- Metadata Management

Data Lineage

Tracks where data originates, how it changes, and where it is consumed.

Benefits

- Easier debugging
- Impact analysis
- Regulatory compliance
- Greater trust in analytics



The background of the image consists of numerous thin, light gray wavy lines that flow across the frame from left to right, creating a sense of movement and depth. These lines are of varying frequencies and amplitudes, giving the background a textured, organic feel.

THANK YOU